Dheeran E. Wiggins

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Education

In Progress **BS Mathematics**, *University of Illinois Urbana-Champaign*
2023 **HS Diploma**, *Illinois Mathematics and Science Academy*

Awards and Fellowships

2026 **Joseph Landin Memorial Scholarship**, *University of Illinois*, (Mathematics Award)
2026 **Undergraduate Research Award**, *Institute for Quantum Computing*, (funded by Mitacs)
2025 **MathQuantum Fellowship**, *University of Maryland*, (funded by NSF DMS-2231533)

Graduate Coursework

- Hilbert Spaces
- Topological Field Theories in 2D
- Representation Theory
- Abstract Algebra I, II
- Geometric Group Theory
- Differentiable Manifolds I
- Homotopy Theory
- Rep-theoretic Methods in QIT
- Algebraic Topology I, II
- General Topology
- Real Analysis

Teaching

Fal 2026 **Quantum Dynamics in One Dimension**, *University of Illinois*
IBL project designed for students in Math 231: Calculus II (course page)

Fal 2026 **Algebra: Preparation and Abstraction**, *University of Illinois*
IBL project designed for James Scholar Honors students in Math 405: Teacher's Course

Fal 2026 **A Typesetting Tutorial for Math in LaTeX**, *University of Illinois*
Tutorial lecture for Math 405: Teacher's Course (course page)

Fal 2026 **Math 231: Calculus II, Discussion BDC**, *University of Illinois*

Jul 2026 **Topological Field Theories in Quantum Computing**, *University of Maryland*
Invited lecture for MathQuantum 2026 High School Fellowship (slides)

Spr 2026 **Three Solutions to the Basel Problem**, *University of Illinois*
IBL project designed for James Scholar Honors students in Math 231: Calculus II (course page)

Spr 2026 **Math 231: Calculus II, Discussion ADP**, *University of Illinois*

Jan 2026 **Monoidal Categories for Quantum Theory**, *Illinois Math and Science Academy*
Minicourse (5 days) for Intersession 2026 (course page)

Jan 2026 **Intro to Quantum Computing and Algorithms**, *Illinois Math and Science Academy*
with Anastasia Perry. Minicourse (5 days) for Intersession 2026

Fal 2025 **Math 231: Calculus II, Discussion BDD**, *University of Illinois*

Sum 2025 **Hidden Subgroups and Quantum Computation**, *Illinois Math and Science Academy*
Minicourse for Summer 2025 (course page)

Jul 2025 **Quantum Channels and Error Correction**, *University of Maryland*
Invited Lecture for MathQuantum 2025 High School Fellowship (slides)

Jan 2025 **Intro to Mathematical Quantum Error Correction**, *Illinois Math and Science Academy*
Minicourse (5 days) for Intersession 2025 (course page)

Learning

Nov 2026 **NE Conference on Categorical Methods**, *CMSA, Harvard University*, Cambridge, MA

Fal 2026 **Defects and quiche**, *University of Illinois*, Champaign, IL
Reading course with Charles Rezk.

Jun 2026 **Summer School on Homotopy Colimits**, *University of Regina*, Regina, Canada

May 2026 **Atlantic TQFT Spring School**, *Dalhousie University*, Halifax, Canada

May 2026 **UG School on Experimental QIP**, *Institute for Quantum Computing*, Waterloo, Canada

Spr 2026 **Modern and classical homotopy theory**, *University of Illinois*, Champaign, IL
Reading course with Jeremiah Heller.

Spr 2026 **Tensor categories**, *University of Illinois*, Champaign, IL
Reading course with Charles Rezk.

AlY 2025 **Groups of operators and infinite tensor products**, *University of Illinois*, Champaign, IL
Independent study with Igor Mineyev.

Fal 2025 **Monoidal, enriched, and dagger categories**, *University of Illinois*, Champaign, IL
Reading course with Charles Rezk.

Aug 2025 **Summer School on Geometry**, *Universiteit Utrecht*, Utrecht, Netherlands

Spr 2025 **Rudiments of von Neumann algebras**, *University of Illinois*, Champaign, IL
Reading seminar with Jihong Cai and Roy Araiza.

Jul 2023 **Summer School on Condensed Matter Physics**, *Princeton University*, Princeton, NJ

Talks

May 2026 **Some (semi)simple things about 2D unitary TQFT with defects**, *MathQuantum Annual Symposium, University of Maryland*, (Invited talk)

Apr 2026 **Unitary field theories with defects in two dimensions**, *Illinois Undergraduate Research Symposium, University of Illinois*, Champaign, IL
Talk advised by James Pascaleff.

Mar 2026 **Locality via (co)limits in an infinite spin chain**, *Rose-Hulman Undergraduate Mathematics Conference, Rose-Hulman Institute of Technology*, Terre-Haute, IN
with Igor Mineyev.

Mar 2026 **Stabilizer formalism in ∞-qubit systems**, *APS Global Physics Summit*, Denver, CO
with Igor Mineyev.

Apr 2025 **A model for the infinite tensor product of groups: Insights from quantum information**, *Illinois Undergraduate Research Symposium, University of Illinois*, Champaign, IL
Talk advised by Igor Mineyev.

Mar 2025 **On noncommutative graphs and Poulin's stabilizer formalism**, *Rose-Hulman Undergraduate Mathematics Conference, Rose-Hulman Institute of Technology*, Terre-Haute, IN
Talk advised by Roy Araiza.

May 2024 **Winter Spaces and the Stabilizer Formalism**, *Illinois Mathematics Lab Open House, University of Illinois*, Champaign, IL
with Anderson et al. Poster advised by Roy Araiza.

Apr 2024 **Modeling a Viral Epidemic With a Concurrent "Misinfodemic"**, *Research Institute Spring Scientific Research Poster Reception*, Evanston, IL
Poster advised by Anthony Solomonides.

Apr 2024 **Modeling a Viral Epidemic With a Concurrent "Misinfodemic"**, *Illinois Undergraduate Research Symposium, University of Illinois*, Champaign, IL
Talk advised by Anthony Solomonides.

Apr 2024 **Winter Spaces and the Stabilizer Formalism**, *Illinois Quantum Information Science & Technology (IQUIST) World Quantum Day, University of Illinois*, Champaign, IL
with Anderson et al. Poster advised by Roy Araiza.

Apr 2023 **Identification and estimated yield of background events in a search for a doubly charged Higgs boson at CMS**, *APS April Meeting*, Minneapolis, MN
with Anne et al. Poster advised by Peter Dong.

Apr 2023 **Identification of optimal production channels for dark photon searches**, *APS April Meeting*, Minneapolis, MN
with Anne et al. Poster advised by Peter Dong.

Apr 2023 **Unidirectional build architecture: Refactoring a HEP data collection codebase**, *Annual IMSAloquium, Illinois Math and Science Academy*, Aurora, IL
Talk advised by Peter Dong.

Apr 2022 **An investigation of triboson decays into four-lepton final states**, *Annual IMSAloquium, Illinois Math and Science Academy*, Aurora, IL
with George Bayliss and Jesus Fileto. Talk advised by Peter Dong.

Research Experiences

2026 **Researcher**, *Institute for Quantum Computing, University of Waterloo*
in Quantum Channels. Advised by Graeme Smith.

2024 – 2025 **Research Scholar**, *Illinois Mathematics Lab, University of Illinois*
in Operator Systems. Advised by Roy Araiza.

2022 – 2024 **Undergraduate Research Fellow**, *Endeavor Health Research Institute*
in Mathematical Epidemiology and Dynamics. Advised by Anthony Solomonides.

2023 **Research Assistant**, *Northwestern University*
in Biomedical Engineering. Advised by Jules Dewald and Hongchul Sohn.

2022 – 2023 **Research Collaborator**, *University of Chicago*
in COVID-19 Outcomes Data Analysis. Under Julian Solway.

2021 – 2023 **Student Researcher**, *Fermilab*
in Dark Photon and Doubly Charged Higgs Searches. Advised by Peter Dong.

Relevant Skills

Symbolic SageMath, Maple, Mathematica
Numerical MATLAB, Scilab, GNU Octave
Programming Python, C++, Java
Markup LaTeX, HTML, CSS
Languages English *fluent*, Tamil *spoken*, Russian *intermediate*, Spanish *elementary*